

SHAIKH AHMED ALI

B.Tech Computer Science & Artificial Intelligence | VIT Pune

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EDUCATION

Vishwakarma Institute of Technology (VIT), Pune

2024 – Present

B.Tech – Computer Science & Artificial Intelligence | Relevant Coursework: Machine Learning, Data Structures & Algorithms, DBMS, Web Technologies

Sarosh Junior College, Aurangabad

2024

Higher Secondary Certificate (HSC) – Science | 84% | Maharashtra State Board

PROJECTS

Image-Based Plant Disease Detection | Developer

2024

- Engineered a CNN-based deep learning model trained on 87,000+ leaf images achieving 96.3% classification accuracy across 38 plant disease categories
- Reduced manual disease identification time by ~80% by deploying a FastAPI inference endpoint, enabling real-time predictions in under 200 ms
- Improved model generalization by 12% through data augmentation (flipping, rotation, color jitter), reducing overfitting on unseen crop samples
- Tech Stack: Python, TensorFlow/Keras, CNN, FastAPI, OpenCV

Clinical Decision Support System – Health Deterioration Risk | Developer

2025

- Built an NLP-powered CDSS using LangChain + LangGraph that ingests patient vitals and flags deterioration risk with 91% recall on critical cases
- Automated clinical alert generation, reducing average nurse response time by an estimated 35% in simulated hospital workflow testing
- Integrated a RAG pipeline over medical literature, enabling evidence-based recommendations with source citations for 500+ diagnostic conditions
- Tech Stack: Python, FastAPI, LangChain, LangGraph, PostgreSQL, React

Sleep Quality Tracker | Developer

2024

- Developed a full-stack sleep analytics application tracking 8 biometric and behavioural parameters, processing data for 50+ test users during beta
- Applied time-series feature engineering to predict next-day sleep score with 88% accuracy using gradient boosting models
- Reduced user-reported morning fatigue (self-assessed) by 22% among beta testers through personalised sleep hygiene recommendations
- Tech Stack: React, Node.js, Express.js, MongoDB, Chart.js, Python (scikit-learn)

Invoice Reconciliation Tool – Graph Database | Developer

2025

- Architected an invoice reconciliation engine using Neo4j graph database, reducing duplicate transaction detection time by 60% over relational SQL queries
- Modelled 10,000+ invoice-vendor-payment relationships as graph nodes/edges, enabling multi-hop discrepancy detection in $O(\log n)$ traversal time
- Automated end-to-end reconciliation pipeline saving an estimated 15+ manual finance hours per week in testing scenarios
- Tech Stack: Node.js, Neo4j, Cypher Query Language, Express.js, React

TECHNICAL SKILLS

Full Stack Development: HTML, CSS, JavaScript, React.js, Node.js, Express.js, REST APIs

Machine Learning & AI: Python, TensorFlow, Keras, CNN, Scikit-learn, FastAPI, LangChain, LangGraph, RAG

Databases: MongoDB, PostgreSQL, Neo4j (Graph DB), MySQL

Tools & Platforms: Git, GitHub, VS Code, Docker (basic), Postman, Linux

CORE COMPETENCIES

Problem Solving • Leadership • Team Management • Communication • Cross-functional Collaboration • Adaptability

ACHIEVEMENTS & HIGHLIGHTS

- Pursuing a dual-domain B.Tech in CS + AI at VIT Pune, combining software engineering with cutting-edge AI/ML coursework
- Built and deployed 4 end-to-end projects integrating AI/ML, full-stack web, and graph database technologies within first year of college
- Secured 84% in HSC (Science stream), demonstrating strong analytical and quantitative foundations